

DPEN022: Enabling Mathematics B

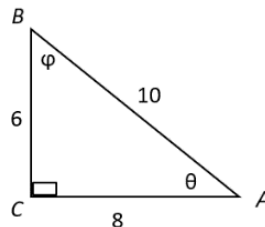
Class Questions Week 1

Week 1A

Trigonometry - Right-Angle Triangles

1. In the right-angled triangle, ABC . Find the following trigonometric ratios.

- (i) $\sin \varphi$
- (ii) $\tan \theta$
- (iii) $\operatorname{cosec} \varphi$
- (iv) $\cot \theta$



2. In a right-angle triangle $\sin A = \frac{5}{13}$, find the following trigonometric ratios.

- (i) $\cos A$
- (ii) $\tan A$
- (iii) $\operatorname{cosec} A$

3. In a right-angle triangle $\cos B = \frac{3}{4}$, find the following trigonometric ratios.

- (i) $\sin B$
- (ii) $\tan B$
- (iii) $\sec B$

4. In a right-angle triangle $\cot C = \frac{1}{3}$, find the following trigonometric ratios.

- (i) $\sin C$
- (ii) $\cos C$
- (iii) $\tan C$

5. Find correct to three decimal places

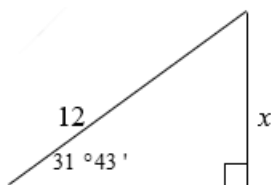
- (i) $\sin 39^\circ 25'$
- (ii) $\cos 45^\circ 50'$
- (iii) $\cot 68.5^\circ$

6. Find the size of acute angle θ in degrees and minutes given the following

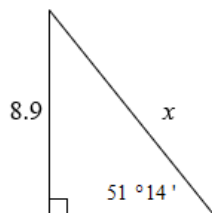
- (i) $\sin \theta = 0.298$
- (ii) $\tan \theta = 0.683$
- (iii) $\operatorname{cosec} \theta = 1.827$

7. Find the value of x :

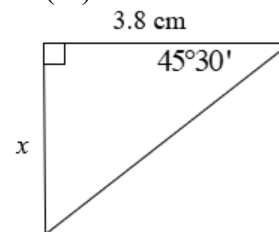
(i)



(ii)

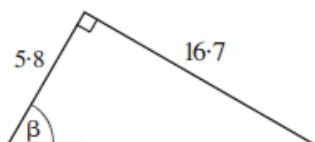


(iii)

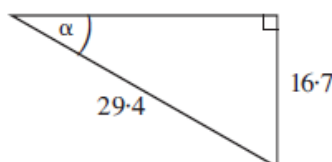


8. Find the size of the angle marked to the nearest degree.

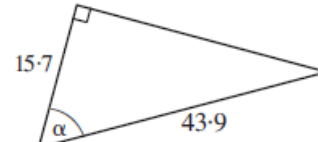
(i)



(ii)



(iii)



9. (i) In a right-angle triangle the length of one side is 6 and the length of the hypotenuse is 10. Find the size of the smallest angle in the triangle, correct to one decimal place.
- (ii) In a right-angle triangle ABC, $a = 7\text{cm}$, $b = 24\text{cm}$ and $c = 25\text{cm}$. Use this information to find the size of angle A and angle B, correct to one decimal place.

Trigonometry - Complementary Angles

- Find the value of x if
 - $\sin x^\circ = \cos 50^\circ$
 - $\sin 80^\circ = \cos(90 - 2x)^\circ$
 - $\tan 80 = \cot(2x)^\circ$
 - $\cos x = \sin(x + 20)^\circ$
- Simplify the following:
 - $\sin x + \cos(90^\circ - x) - 2 \sin x$
 - $\sec \theta - \operatorname{cosec}(90^\circ - \theta)$
- Without using a calculator, simplify the following:
 - $\frac{\sin 50^\circ}{\cos 40^\circ}$
 - $\frac{\tan 35^\circ}{\cot 55^\circ}$
 - $\sec 20^\circ - \operatorname{cosec} 70^\circ$

Trigonometry - Exact Trigonometric Ratios 30° , 45° , 60°

- Find the *exact* values of the following without using a calculator:
 - $\sin 60^\circ + \cos 60^\circ$
 - $\cos^2 45^\circ + \sin^2 45^\circ$
 - $2 \sec 60^\circ$
 - $\operatorname{cosec} 45^\circ$
- If $0^\circ \leq \theta^\circ \leq 90^\circ$, find all values of θ° which satisfy each of the following equations:
 - $\sin \theta = \frac{1}{\sqrt{2}}$
 - $\cos \theta = \frac{1}{\sqrt{2}}$
 - $\tan \theta = 1$
 - $\cot \theta = \frac{1}{\sqrt{3}}$
 - $\sec \theta = 2$
 - $\operatorname{cosec} \theta = \sqrt{2}$

Week 1B

Trigonometry - Angles of Any Magnitude

- In which quadrant (1^{st} , 2^{nd} , 3^{rd} , 4^{th}), will the following angles be found?
 - 110°
 - 210°
 - 39°
 - 308°
 - 281°
 - 97°
 - 456°
 - -25°
 - -135°
- State whether the following ratios will be positive or negative:
 - $\sin 170^\circ$
 - $\cos 280^\circ$
 - $\tan 60^\circ$
 - $\tan 125^\circ$
 - $\sin 56^\circ$
 - $\cos 226^\circ$
 - $\cot 13^\circ$
 - $\operatorname{cosec} 320^\circ$
 - $\sec 165^\circ$
 - $\cos(-45^\circ)$
 - $\sin 370^\circ$
 - $\sin(-125^\circ)$
- Complete the table by determining the sign of the trigonometric ratios.

Angle θ	Quadrant	$\sin \theta$	$\cos \theta$	$\tan \theta$
120°	2	+	-	-
300°				
240°				
460°				

4. In which quadrant would angle θ be located if:

(i) $\sin \theta > 0$, $\cos \theta < 0$

(ii) $\tan \theta > 0$, $\sin \theta < 0$

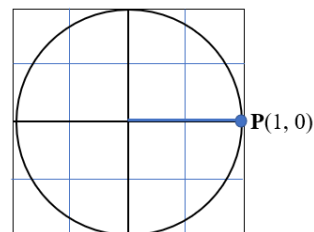
(iii) $\cos \theta > 0$, $\sin \theta > 0$

(iv) $\tan \theta < 0$, $\cos \theta > 0$

Trigonometry - Exact Trigonometric Ratios 0° , 90° , 180° , 270° 360°

1. On the unit circle, the point P on the circumference corresponds to an angle of 0° and has coordinates of $(1, 0)$. $\rightarrow$

Complete the table below by showing the coordinates of points on the circumference corresponding to specified angles. Then use the coordinates to identify the trigonometric ratios.



Angle θ	Coordinates	$\sin \theta$	$\cos \theta$
0°	$(1, 0)$	0	1
90°			
180°			
270°			
360°			

2. Using the exact results of the trigonometric functions (without using a calculator), evaluate the following (if possible):

(i) $4 \cos 0^\circ$

(ii) $\tan 180^\circ + \tan 0^\circ$

(iii) $\sin 90^\circ + \sin 180^\circ$

(iv) $\cos 540^\circ$

(v) $\cos 180^\circ \sin 270^\circ$

(vi) $2 \cos 90^\circ \sin 90^\circ$

(vii) $\sin^2 270^\circ + \cos^2 270^\circ$

(viii) $\operatorname{cosec} 180^\circ$

(ix) $3 \sec 0^\circ - \tan 180^\circ$

Trigonometry – Angles of any Magnitude

1. Find the exact values of the following:

(i) $\cos 150^\circ$

(ii) $\tan 150^\circ$

(iii) $\sin 240^\circ$

(iv) $\cos 315^\circ$

(v) $\sin -120^\circ$

(vi) $\tan 570^\circ$

(vii) $\tan -45^\circ$

(viii) $\cos 210^\circ$

(ix) $\sin -300^\circ$

(x) $\operatorname{cosec} 300^\circ$

(xi) $\cot 135^\circ$

(xii) $\sec 225^\circ$

Trigonometry - Solving Trigonometric Equations in Degrees

1. If $0^\circ \leq \theta \leq 360^\circ$ find all values of θ which satisfy each of the following equations:

(i) $\sin \theta = \frac{1}{\sqrt{2}}$

(ii) $\cos \theta = \frac{1}{\sqrt{2}}$

(iii) $\tan \theta = -1$

(iv) $\sin \theta = -\frac{1}{2}$

(v) $\tan \theta = \sqrt{3}$

(vi) $\cot \theta = \frac{1}{\sqrt{3}}$

(vii) $\sec \theta = -2$

(viii) $\operatorname{cosec} \theta = \sqrt{2}$

(ix) $2 \cos \theta = -1$

(x) $\tan^2 x = 1$

(xi) $\sqrt{3} \tan \theta = 1$

(xii) $\cos^2 \theta - \frac{1}{4} = 0$

2. If $0^\circ \leq \theta \leq 360^\circ$ find all values of θ which satisfy each of the following equations.

Give the angles correct to the nearest minute.

(i) $\sin \theta = 0.4827$

(ii) $\cos \theta = -0.8731$

(iii) $\operatorname{cosec} \theta = 3.6771$

Week 1C

Trigonometry - Radian Measure

1. Express the following in radians, leaving your answers in terms of π :

(i) 120° (ii) 18° (iii) -225°

2. Express the following in degrees:

(i) $\frac{17\pi}{12}$ (ii) 1.2 (iii) -3.15

3. State the quadrant in which each of the following angles lie:

(i) $\frac{\pi}{3}$ (ii) $\frac{3\pi}{4}$ (iii) $\frac{4\pi}{3}$ (iv) $\frac{10\pi}{6}$
(v) $\frac{11\pi}{12}$ (vi) $\frac{13\pi}{6}$ (vii) $\frac{7\pi}{8}$ (viii) $\frac{\pi}{12}$

Trigonometry – Angles of any Magnitude in Radians

4. Find an exact value for each ratio:

(i) $\tan \pi$ (ii) $\sin \frac{3\pi}{2}$ (iii) $\cos \frac{\pi}{4}$ (iv) $\sin \frac{\pi}{6}$
(v) $\cos \frac{2\pi}{3}$ (vi) $\tan \frac{5\pi}{4}$ (vii) $\sin \pi$ (viii) $\sin \frac{\pi}{2}$
(ix) $\tan \frac{3\pi}{4}$ (x) $\sin \frac{2\pi}{3}$ (xi) $\cos \frac{5\pi}{3}$ (xii) $\sin \frac{4\pi}{3}$
(xiii) $\sin \frac{5\pi}{6}$ (xiv) $\cos \frac{5\pi}{4}$ (xv) $\tan \frac{4\pi}{3}$ (xvi) $\cos(-\frac{4\pi}{3})$

Trigonometry - Solving Trigonometric Equations in Radians

1. If $0 \leq x \leq 2\pi$ find all exact values of x which satisfy each of the following equations:

(i) $\sin x = -\frac{1}{\sqrt{2}}$ (ii) $\cos x = \frac{1}{\sqrt{2}}$ (iii) $\tan x = -1$
(iv) $\sin x = \frac{1}{2}$ (v) $\tan x = \sqrt{3}$ (vi) $\cot x = \frac{1}{\sqrt{3}}$
(vii) $\sec x = -2$ (viii) $\operatorname{cosec} x = \sqrt{2}$ (ix) $2 \cos x = -1$
(x) $\tan^2 x = 1$ (xi) $\sqrt{3} \tan x = 1$ (xii) $\cos^2 x - \frac{1}{4} = 0$

2. If $0 \leq x \leq 2\pi$, find all values of x which satisfy each of the following equations.

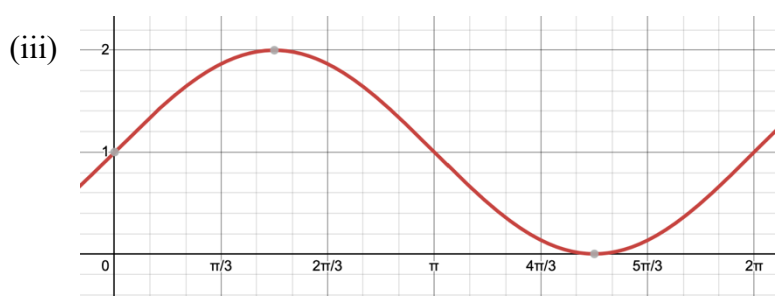
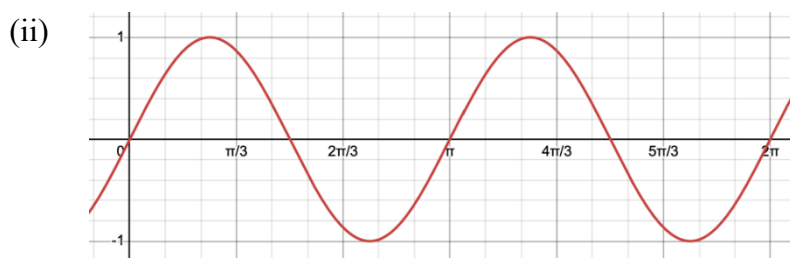
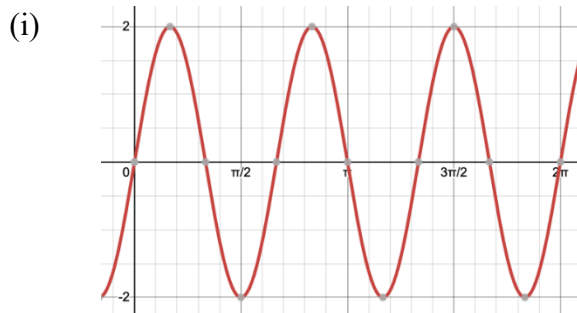
(i) $\sin x = 0.4827$ (ii) $\operatorname{cosec} x = 3.6771$

Week 1D

Trigonometry - Graphs of Trigonometric Functions

1.
 - (i) The domain of the function $y = \cos x$ is?
 - (ii) The range of the function $y = \cos x$ is?
 - (iii) The graph of $y = \cos x$ has a period _____ and amplitude _____.
 - (iv) The graph of $y = a \cos nx$ has a period _____ and amplitude _____.
2. Determine the amplitude and period for each of the following functions.
 - (i) $y = \cos 2x$
 - (ii) $y = 2 \cos x$
 - (iii) $y = \frac{1}{4} \cos 4x$
 - (iv) $y = -4 \cos \frac{x}{4}$
 - (v) $y = 3 \cos 2x$
 - (vi) $y = 2 - 2 \cos x$
 - (vii) $y = -0.5 \cos 2\pi x$
3. Sketch the graph of the following functions.
 - (i) $y = 3 \cos x$
 - (ii) $y = \cos 3x$
 - (iii) $y = 2 \cos 2x$
 - (iv) $y = \frac{1}{2} \cos \frac{x}{4}$
4.
 - (i) The domain of the function $y = \sin x$ is?
 - (ii) The range of the function $y = \sin x$ is?
 - (iii) The graph of $y = \sin x$ has a period _____ and amplitude _____.
 - (iv) The graph of $y = a \sin nx$ has a period _____ and amplitude _____.
5. Determine the amplitude and period for each of the following functions.
 - (i) $y = \sin 4x$
 - (ii) $y = 4 \sin x$
 - (iii) $y = \frac{1}{2} \sin 2x$
 - (iv) $y = -2 \sin \frac{x}{4}$
 - (v) $y = 2 \sin x$
 - (vi) $y = 3 \sin 4x$
 - (vii) $y = 4 \sin \pi x$
6. Sketch the graph of the following functions.
 - (i) $y = 2 \sin x$
 - (ii) $y = \sin 2x$
 - (iii) $y = \sin \pi x$
 - (iv) $y = 4 \sin \frac{x}{2}$
7.
 - (i) The domain of the function $y = \tan x$ is?
 - (ii) The range of the function $y = \tan x$ is?
 - (iii) The graph of $y = \tan x$ has a period _____ and amplitude _____.
 - (iv) The graph of $y = a \tan nx$ has a period _____ and amplitude _____.
8. Determine the amplitude and period for each of the following functions.
 - (i) $y = \tan 4x$
 - (ii) $y = 3 \tan x$
 - (iii) $y = -\tan \pi x$
 - (iv) $y = 2 \tan \frac{x}{2}$
 - (v) $y = \tan 2x$
 - (vi) $y = 1 - 4 \tan 3x$
9. Sketch the graph of the following functions.
 - (i) $y = \tan x$
 - (ii) $y = \tan 2x$
 - (iii) $y = 3 \tan x$
 - (iv) $y = -\tan x$
10. Determine the number of cycles in interval $0 \leq x \leq 2\pi$ and the period for the following functions.
 - (i) $y = 10 \sin x$
 - (ii) $y = 1 + 8 \sin 4x$
 - (iii) $y = 1 - \sin \frac{x}{2}$
 - (iv) $y = -2 \cos \left(\frac{x}{3} \right) + 2$

11. For each of the graphs below, determine the amplitude and number of cycles in interval $0 \leq x \leq 2\pi$. Then write the sine equation for each curve.



12. The graph of $y = \sin x$ undergoes the following transformations:

- vertically stretched so the amplitude is 3.
- horizontally stretched so that the period is 4π .
- shifted vertically down by 2 units

Write the equation for the resulting sine function.

13. The graph of $y = \sin x$ undergoes the following transformations:

- vertically stretched so the amplitude is 4.
- horizontally compressed so that the period is π .
- shifted vertically up by 3 units

Write the equation for the resulting sine function.

14. Write the equation for $g(x)$ using the following information.

Relative to the parent function $y = \sin x$, the graph of $g(x)$ is

- shifted up by 1 unit
- shifted to the left by $\frac{\pi}{6}$ units.
- vertically stretched so that the amplitude is 2.